Worksheet I (Game Theory)

COMP6203 - Intelligent Agents

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Question 1

Question

Let $\Omega = \{\omega_1, \ldots, \omega_8\}$ be a set of outcomes. Suppose that an agent has the following preference ordering over Ω:

$$\omega_1 \succ \omega_2 \succeq \omega_3 \succeq \omega_4 \succ \omega_5 \succeq \omega_6 \succeq \omega_7 \succeq \omega_8.$$

Which of the following utility functions $u : \Omega \to \mathbb{R}$ represent the above preference?

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$$u(x) = \frac{1}{i}, \text{ for } x = \omega_i$$

YES: $u(\omega_1) = 1, \quad u(\omega_2) = \frac{1}{2}, \quad u(\omega_3) = \frac{1}{3}, \quad \ldots$

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YES: $u(\omega_1) = 1, \quad u(\omega_2) = \frac{1}{2}, \quad u(\omega_3) = \frac{1}{6}, \quad \ldots$

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$$u(x) = \begin{cases} 7 & \text{if } x = \omega_1, \\ 5 & \text{if } x = \omega_i,\ 2 \leq i \leq 4 \\ 3 & \text{if } x = \omega_i,\ 5 \leq i \leq 8 \end{cases}$$

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YES: $u(\omega_1) = 7, u(\omega_2) = u(\omega_3) = u(\omega_4) = 5, \quad \ldots$

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$$u(x) = \begin{cases} 3 & \text{if } x = \omega_1 \\ 5 & \text{if } x = \omega_i,\ 2 \leq i \leq 4 \\ 7 & \text{if } x = \omega_i,\ 5 \leq i \leq 8 \end{cases}$$

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NO: $u(\omega_1) = 3, u(\omega_2) = u(\omega_3) = u(\omega_4) = 5, \quad \ldots$

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$$u(x) = 10 \text{ for all } x \in \Omega$$

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NO: obvious

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$$u(x) = \sum_{j=1}^{i} \frac{1}{j}, \text{ for } x = \omega_i$$

NO: $u(\omega_1) = 1, \quad u(\omega_2) = \frac{3}{2}, \quad u(\omega_3) = \frac{11}{6}, \quad \ldots$

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$$u(x) = \prod_{j=1}^{i}(-j), \text{ for } x = \omega_i$$

NO: $u(\omega_1) = -1, \quad u(\omega_2) = 2, \quad u(\omega_3) = -6, \quad \ldots$

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YES: $u(\omega_1) = -1, \quad u(\omega_2) = -3, \quad u(\omega_3) = -6, \quad \ldots$

Question

Consider the following strategic-form game

		Player 2				
		F	G	H	I	J
Player 1	A	5, 7	4, 1	4, 4	3, 7	3, 4
	B	4, 4	5, 3	5, 8	8, 1	3, 4
	C	1, 6	3, 3	3, 5	2, 6	1, 4
	D	4, 3	6, 1	8, 2	10, 3	2, 3
	E	5, 3	5, 1	4, 2	8, 3	3, 3

1. *Which of the above strategies are weakly dominated (by some other strategy)?*
2. *Compute the pure Nash equilibria of the game after eliminating weakly dominated strategies.*
3. *Compute the whole set of pure Nash equilibria.*

Question 2

Weakly dominated strategies:

		Player 2				
		F	G	H	I	J
Player 1	A	5, 7	4, 1	4, 4	3, 7	3, 4
	B	4, 4	5, 3	5, 8	8, 1	3, 4
	C	1, 6	3, 3	3, 5	2, 6	1, 4
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1. A is weakly dominated by E

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1. A is weakly dominated by E
2. C is strictly (and therefore weakly) dominated by all other strategies

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Weakly dominated strategies:

		Player 2				
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Player 1	A	5, 7	4, 1	4, 4	3, 7	3, 4
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3. G is strictly (and therefore weakly) dominated by F

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1. A is weakly dominated by E
2. C is strictly (and therefore weakly) dominated by all other strategies
3. G is strictly (and therefore weakly) dominated by F
4. I is weakly dominated by F

Question 2

Weakly dominated strategies:

		Player 2				
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Player 1	A	5, 7	4, 1	4, 4	3, 7	3, 4
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1. A is weakly dominated by E
2. C is strictly (and therefore weakly) dominated by all other strategies
3. G is strictly (and therefore weakly) dominated by F
4. I is weakly dominated by F
5. J is weakly dominated by F

Question 2

We can follow different orders of elimination...let's start with A

Player 1 \ Player 2	F	G	H	I	J
A					
B	4, 4	5, 3	5, 8	8, 1	3, 4
C	1, 6	3, 3	3, 5	2, 6	1, 4
D	4, 3	6, 1	8, 2	10, 3	2, 3
E	5, 3	5, 1	4, 2	8, 3	3, 3

We eliminate G

		Player 2				
		F	G	H	I	J
Player 1	A					
	B	4, 4		5, 8	8, 1	3, 4
	C	1, 6		3, 5	2, 6	1, 4
	D	4, 3		8, 2	10, 3	2, 3
	E	5, 3		4, 2	8, 3	3, 3

We eliminate C

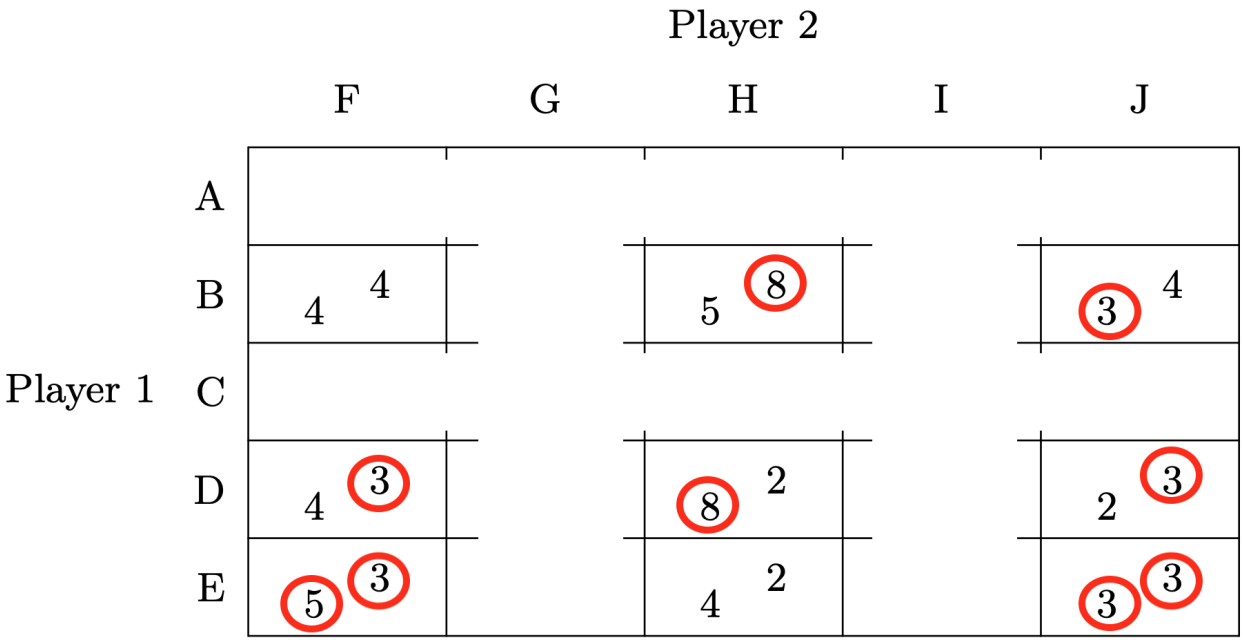
		Player 2				
		F	G	H	I	J
Player 1	A					
	B	4, 4		5, 8	8, 1	3, 4
	C					
	D	4, 3		8, 2	10, 3	2, 3
	E	5, 3		4, 2	8, 3	3, 3

We eliminate I

Player 1 \ Player 2	F	G	H	I	J
A					
B	4, 4		5, 8		3, 4
C					
D	4, 3		8, 2		2, 3
E	5, 3		4, 2		3, 3

Question 2

Pure Nash equilibria in the game obtained after elimination: $(E, F), (E, J)$



Question 2

Pure Nash equilibria for the whole game: $(A, F), (D, I), (E, F), (E, J)$

Player 1 \ Player 2	F	G	H	I	J
A	(5, 7)	4, 1	4, 4	3, (7)	(3), 4
B	4, 4	5, 3	5, (8)	8, 1	(3), 4
C	1, (6)	3, 3	3, 5	2, (6)	1, 4
D	4, (3)	(6), 1	(8), 2	(10), (3)	2, (3)
E	(5), (3)	5, 1	4, 2	8, (3)	(3), (3)

Question

Consider the following game. One night both your house and your neighbour's get burglarized. The burglar steals exactly the same items from both houses. You and your neighbour file a claim with the same insurance company asking for compensation for the burglary. The insurance company contacts you and your neighbour separately and asks each of you to estimate the value of what was stolen. You can claim a maximum of £1000 and a minimum of £50, but how much money you receive depends on how honest you are with your estimation. If both you and your neighbour ask for the same amount of money, you both get exactly that amount. If you ask for different amounts, you will both receive the smaller amount with a penalty/reward of £50: so, if you ask for x and y with $x < y$, the person asking for x will receive £$(x + 50)$, while the person asking for y will receive £$(x - 50)$. Suppose that at no point you can communicate with your neighbour and that each of you can claim any amount of money from

$$M = \{£50, £60, £70, \ldots, £980, £990, £1000\}.$$

You can see this as a strategic-form game where you and your neighbours are the players.

What is the set of all pure strategy Nash equilibria?

Question 3

- This is a game similar to the traveler's dilemma.
- If both players choose £1000, they both get £1000
- However, one of them (say player 1) benefits from deviating to £990, which increases their payoff but decreases the payoff for player 2.
- Player 2 then is incentivised to match player 1 and play £990.
- Again, player 1 can deviate by asking for a lower amount, etc.
- The process keeps repeating...
- This shows that the only equilibrium is when players both choose £50.

Question 3

- Formally, take any

$$x, y \in \{£60, £70, \ldots, £980, £990, £1000\},$$

  (where x is player 1's strategy and y is player 2's strategy).

- If $x = y$, then for any $x' = x - £10$, for player 1, $u_1(x', y) > u_1(x, y)$ (similar for player 2).

- If $x > y$, then for player 1 $u_1(y, y) > u_1(x, y)$. Similar for player 2 if $x < y$.

- $(£50, £50)$ is an equilibrium because each player will receive £0 by unilaterally deviating.

Question

Consider the following strategic-form game with two players P1 and P2. Let $k > 0$ be any natural number. Each player has the same strategy set

$$S = \left\{0, \frac{1}{k}, \frac{2}{k}, \ldots, \frac{k-1}{k}, 1\right\}.$$

The following are the utility functions for P1 and P2, respectively:

$$u_1(x_1, x_2) = |x_1 - x_2|, \qquad u_2(x_1, x_2) = \sqrt[4]{1 - (|x_1 - x_2|)^4}.$$

Compute the set of equilibria of the game.

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Compute the set of equilibria of the game.

Take any strategy combination (x_1, x_2). P1 can always try to maximise the distance from P2 by deviating to 0 or 1 (if possible). On the contrary, P2 can maximise their utility by matching P1's choice and so can always deviate to x_2 (if possible). So, the set of pure strategy equilibria is $\emptyset$.

Question 4

- Formally, take any
$$x, y \in \left\{0, \frac{1}{k}, \frac{2}{k}, \ldots, \frac{k-1}{k}, 1\right\},$$
(where x is player 1's strategy and y is player 2's strategy).
- If $x = y$, then for any $x' \neq x$, for player 1, $u_1(x', y) > u_1(x, y)$.
- If $x \neq y$, then for $y' \neq x$, $u_2(x, y') > u_2(x, y)$ for player 2.
- So the game contains no pure strategy Nash equilibria.